

Computational Screening of Redox-Active Monomers

A physics-based, multi-objective candidate-selection pipeline

RTFB Materials Research

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Redox potential · Stability · Kinetics · Capacity

The goal: pick good redox molecules for flow batteries

System. A **Merrifield-resin polymer** carrying pendant redox groups. The redox chemistry is *local*, so we model the **monomer as a proxy**: establish we can capture the local chemistry, then add backbone effects.

Selection is multi-objective. Voltage and stability are necessary but not sufficient:

$$\text{Energy density} \propto n \cdot V_{\text{cell}} \cdot \text{solubility}$$

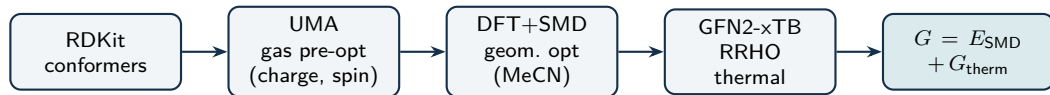
- **redox potential** $\rightarrow$ cell voltage / window
- **stability** $\rightarrow$ cycle life
- **kinetics** (λ), **capacity** (n), cost, crossover

Guiding principle

Energy *differences* with error cancellation compute **reliably**; *absolute* charged-species solvation does **not**.

Build objectives on the cancellation-friendly quantities; treat the rest as labelled proxies or filters.

The pipeline (per redox state)



Level of theory

- opt: r²SCAN/def2-SVP(D) + SMD
- energy: ω B97M-V/def2-TZVP
- referenced to level-matched Fc/Fc⁺

Two hard-won fixes

- gpu4pyscf open-shell Hessian is **broken** → thermal from **GFN2-xTB** RRHO
- cross-points gas-only (do_smd=False), ~2× cheaper

How each axis is computed

axis	method	validation / trust
redox potential	DFT+SMD ΔG ; ω B97M-V/def2-TZVP // r ² SCAN/def2-SVP(D); level-matched Fc ref	OROP exp: MAE 0.58 V, ρ 0.95
thermal G_{therm} stability (disprop.)	GFN2-xTB RRHO (298 K) $\Delta G_{\text{disp}} = F(E_{\text{hi}} - E_{\text{lo}})$ from state G 's	open-shell safe exp spacings: MAE 0.17 eV, ρ 1.0
stability (dimer.) reorg. energy λ	pimer cluster DFT+SMD ΔG λ_i 4-point (Nelsen), gas + λ_o (Marcus/Born, Eq. 13)	flagged (+2 desolvation) D3TaLES λ_i : $R^2=1.0$ ($n=2636$)
reversibility capacity	bond-graph change + solvated EA $n \cdot F/\text{MW}$ (accessible n)	catches dissoc. (CCl_4) exact
solubility	$\Delta G_{\text{solv}} = E_{\text{SMD}} - E_{\text{gas}}$ (neutral)	proxy (relative)
synth. access.	RDKit SA_Score	heuristic proxy

Geometries optimized in SMD(MeCN); conformers via RDKit + UMA pre-opt.

Redox potential: benchmarked vs. experiment (OROP)

Blind test on OROP MeCN potentials
($n = 36$):

	ours	OROP impl.-DFT
MAE (V)	0.58	0.43
signed bias	+0.46	—
Spearman (all)	0.95	—

Absolute accuracy sits at the $\sim$ **0.5–0.9 V** **physics floor** of implicit-solvent DFT — the same for everyone. **Ranking**, not absolute potential, is the usable output.

Ranking must be judged *within* charge class

class	n	MAE	Spearman
cations (+1)	21	0.44	0.885
anions (0)	13	0.77	0.489
anions (filtered)	8	0.53	0.952

Global 0.95 is inflated (charge classes separate).
Cations rank well; anions needed a fix → next slide.

The anion “failure” was category errors, not bad physics

The anion ranking collapse (Spearman 0.49) traced to a few molecules with **no reversible reduction**:

molecule	exp	RMSD _{neu→an}
CCl ₄	-2.57	3.4 Å
CH ₂ Br ₂	-3.12	1.3 Å
(unbound anions)	—	EA _{gas} < 0

Dissociative electron attachment
(CCl₄+e⁻→CCl₃[•]+Cl⁻) — scored against irreversible *peak* potentials.

Reversibility screen

Bound anion (solvated EA > 0) *and* intact (**bond-graph unchanged**, not raw RMSD — which is confounded by tether flexibility).

- filtering → anion Spearman **0.49**→**0.95**
- a *required* screening criterion + a *stability* signal
- all 20 candidate couples: **reversible**

Stability = decomposition free energies

Disproportionation $2R \rightarrow R^+ + R^-$
(atom- & electron-conserving $\rightarrow$ **cancellation**
 $\rightarrow$ **reliable**):

intermediate	ΔG_{disp} (kJ/mol)
TEMPO $\bullet$	+221
AQ semiquinone	+86
viologen MV $^{+\bullet}$	+52

TEMPO — the textbook persistent radical (our validation anchor) — ranks most stable, as it must. Exact from existing energies; no new calculation.

Dimerization $2\text{MV}^{+\bullet} \rightarrow [\text{MV}_2]^{2+}$ (the real viologen failure mode):

$$\Delta G_{\text{dim}} = +0.49 \text{ eV (flagged)}$$

Config sampling: minimum well-defined ($\sigma_{\text{config}} \approx 2 \text{ meV}$).

The uncertainty is the **+2 pimer desolvation** — the continuum's weak spot.

$\Rightarrow$ use dimerization as a **relative** ranking; lean on disproportionation as the trustworthy axis.

Kinetics: reorganization energy $\lambda = \lambda_i + \lambda_o$

Inner-sphere λ_i — solute geometry relaxation, **4-point (Nelsen)**:

$$\lambda_i = [E_O(q_R) - E_O(q_O)] + [E_R(q_O) - E_R(q_R)]$$

gas single points; *identical* to D3TaLES.

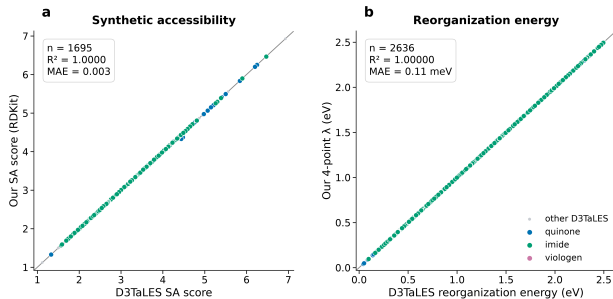
Validated.

Outer-sphere λ_o — solvent reorganization (Marcus/Born, **Sharma Eq. 13**):

$$\lambda_o = \frac{z^2 e^2}{8\pi\epsilon_0 r_{\text{hyd}}} \left(\frac{1}{\epsilon_{\text{op}}} - \frac{1}{\epsilon_r} \right)$$

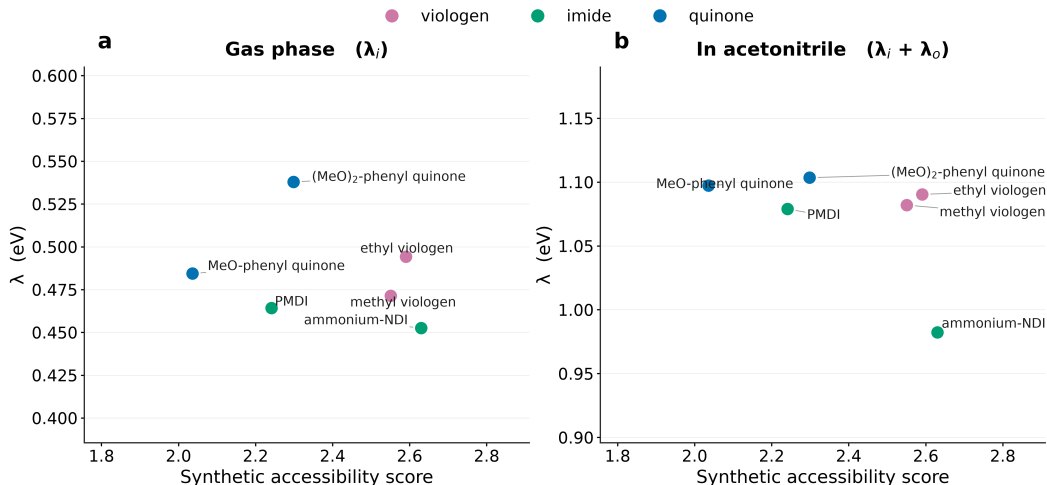
MeCN ($\epsilon_r=37.5$, $\epsilon_{\text{op}}=1.81$); SASA r_{hyd} proxy $\Rightarrow$ **semi-quantitative** ($\sim 2\times$).

$\lambda_{\text{total}} = \lambda_i + \lambda_o$: trust λ_i ; read λ_o as *relative*.



Independently reproduce D3TaLES's SA & inner-sphere λ :
 $R^2 = 1.0$ (SA $n=1695$; λ $n=2636$; quinones & imides highlighted).

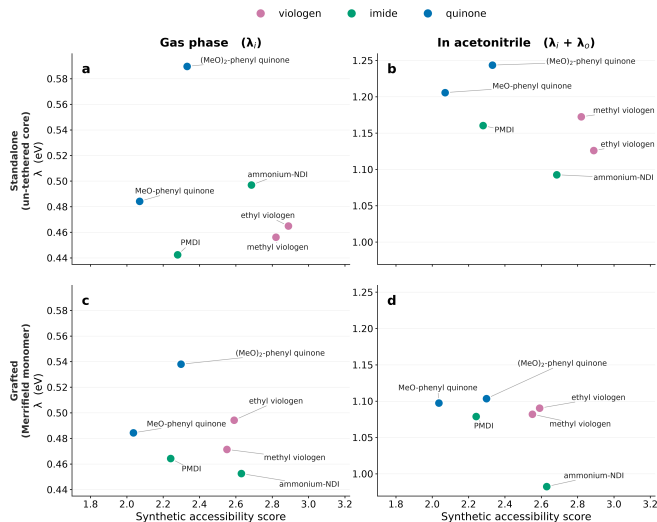
Candidates: SA vs. reorganization energy



one point per molecule (mean over its charge states); note the independent y-zoom per panel — solvation adds ~ 0.5 – 0.6 eV outer-sphere.

All six are *grafted* model-compounds (redox core + 4-methylbenzyl monomer proxy); short class names label them.

Functionalization $\times$ solvation: the 2×2



Rows = **before** (standalone core) vs **after** functionalization (grafted: redox core + 4-methylbenzyl monomer handle).

Cols = gas (λ_i) vs MeCN ($\lambda_i + \lambda_o$).

Functionalization barely moves λ_i — ± 15 – 50 meV, mixed sign:

- viologens: +15 / +29 meV
- PMDI +22; **NDI -44**; quinones 0 / -52

Solvation adds a near-uniform **+0.5–0.6 eV (λ_o)**.

$\Rightarrow$ the **redox core sets λ_i** ; the monomer handle is a small perturbation, so the cheap standalone λ is a good predictor of the grafted value.

What we learned (the durable lessons)

What works

- ranking (within charge class)
- xtb RRHO thermal (open-shell safe)
- disproportionation stability
- reorganization energy λ
- reversibility as filter + stability signal
- capacity (n , MW): exact

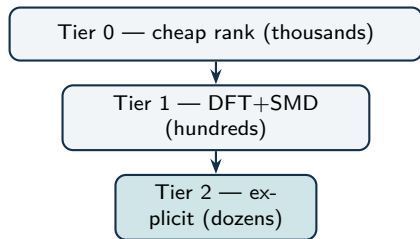
Negative results (saved effort)

- continuum ion-pairing: **refuted** (spacing off by ~ 11 V)
- explicit microsolvation (4 MeCN): **no gain** for viologen
- absolute redox potential: capped ~ 0.5 V
- dimerization absolute: **unreliable** (+2 desolvation)
- solubility / crossover: proxies only

Corroborated by the metabolic- ΔG pipeline: 0.14 eV via isodesmic cancellation $\Rightarrow$
cancellation, not more physics, is the lever.

Putting it together: a tiered, multi-objective search

Tiered funnel (explicit solvation is a finisher, never for thousands):



Pareto scorecard — built correctly

- objectives = **trustworthy** axes only (capacity, λ , disprop. stability, ranked potential)
- proxies (solubility, SA, crossover) = **filters**, not axes
- **reversibility** = hard pre-filter
- split **anolyte** / **catholyte** (potential isn't "more is better")
- σ -**aware** domination (don't rank within error)
- pair front with $n \cdot V$ ·solubility figure of merit

Applying it: candidate scorecard → shortlist

σ -aware Pareto on the **six grafted starting candidates** — all fall in the **anolyte** pool (2e reductions / viologen reductions), all reversible, none gated out:

id	E_{an}	Cap	λ	ΔG_d	FoM	Par.
pmdi	-1.49	160	0.46	73	0.77	—
mophquinone*	-1.05	176	0.48	101	0.59	Y
viologen*	-1.03	194	0.47	52	0.54	Y
ethylviologen	-1.03	185	0.49	50	0.45	—
dmophquinone	-1.01	147	0.54	101	0.33	—
ndi_ammonium	-1.00	114	0.45	29	0.20	—

* = Pareto-optimal ($\Rightarrow$ shortlist: **mophquinone**, **viologen**).

σ -aware matters: pmdi has the *top* FoM (best voltage, -1.49 V) yet is *dropped* from the Pareto front — its 0.44 V edge sits inside $\sigma_E \approx 0.8$ V, so it is σ -dominated by the higher-capacity / more-stable mophquinone & viologen.

$\Rightarrow$ across this tight set voltage is *non-discriminating*; **capacity + disproportionation stability + λ** drive the ranking.

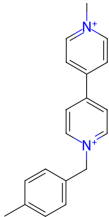
E_{an} [V vs Fc], Cap [mAh/g], λ [eV], ΔG_d (disprop.) [kJ/mol].

Caveats surfaced automatically: the quinones' larger ΔG_{disp} (persistent semiquinone) offsets their lower capacity; viologen carries a flagged dimerization mode (ΔG_{dim} from earlier). All six are anolyte-side.

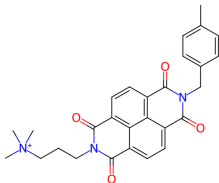
Current focus: six multi-electron starting candidates

Starting candidates (Merrifield-grafted monomers)

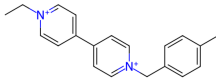
Methyl viologen



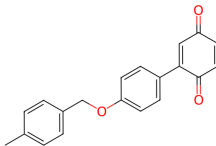
pyridine-multi-e · charge states: +2, +1, 0
Ammonium naphthalene diimide



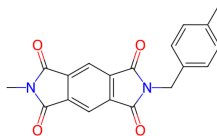
Ethyl viologen



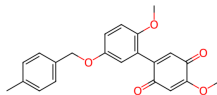
pyridine-multi-e · charge states: +2, +1, 0
Methoxyphenyl-benzoquinone



Pyromellitic diimide (PMDI)



imide (n-type) · charge states: 0, -1, -2
Dimethoxyphenyl-methoxyquinone



Next steps

Done: all axes computed & validated; unified σ -tagged scorecard; σ -aware Pareto engine; shortlist on the six grafted starting candidates (**mophquinone**, **viologen** Pareto-optimal).

1. **Scale up discovery** — run the tiered funnel on a larger design space (functionalized variants + D3TaLES-mined pools), not just the current six.
2. **Harden for scale** — SCF-robustness patch ($\sim 20\%$ of jobs hit gradient non-convergence); needed before screening hundreds.
3. **Broaden benchmarks** — more two-wave molecules for stability; more OROP for the redox ranking, esp. per charge class.
4. **Polymer effects** — backbone/tether constraints & inter-monomer dimerization (where the *relative* dimerization ranking becomes key).
5. **Absolute accuracy** (if needed) — same-charge referencing to remove the per-class redox bias.

Summary

We can **rank** redox candidates and screen **stability**, **kinetics**, and **reversibility** on a physics-based footing — with each axis **honestly labelled** by how far we can trust it.

The σ -aware scorecard & Pareto engine are built and applied. Next: point the funnel at a larger design space.